LAB WORKSHEET

Motherboard Lab

Identify the parts of an ATX motherboard, what plugs into each one, and how the CPU and chipset connect everything.

Key facts

- CPU power: 8-pin **EPS** → VRM → CPU. Main board power: **24-pin ATX**.
- RAM: two sticks in **A2 + B2** for dual channel; the memory controller is in the CPU.
- PCIe x16_1 is wired to the **CPU** (Gen5 ×16 ≈ 63 GB/s). x16_2 is full length but wired **x4** through the chipset (≈ 7.9 GB/s).
- The **chipset** (SATA, USB, M2_2, x1 slots, LAN) shares the **DMI** link to the CPU.
- Per lane: Gen3 ≈ 1, Gen4 ≈ 2, Gen5 ≈ 4 GB/s. A card and slot run at the lowest generation and fewest lanes.
- ARGB = 5 V 3-pin; RGB = 12 V 4-pin. USB 2.0 header = 9-pin; USB 3 = 19-pin.
- Form factors: ATX 305 × 244 mm · Micro-ATX 244 × 244 mm · Mini-ITX 170 × 170 mm.

Tabs

1. **Explore** — hover/click every component.
2. **Identify** — drag 20 labels onto the board.
3. **Where does it plug in?** — fit 13 components.
4. **Form factors** — to-scale overlay and quiz.
5. **Buses & lanes** — CPU vs chipset lanes, DMI, bandwidth calculator.
6. **Quiz** — 12 questions.

Guided steps

Open the socket, EPS/VRM and chipset cards → label the board → fit the CPU, RAM (A2), GPU, 24-pin and EPS → form-factor questions → bottom-slot scenario → Gen4 ×16 calculation → finish the quiz.

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